

# Ricardo (Zhicong) Xie

## Manufacturing / Site Project Intern | CAD-to-Part, Workshop Practice and Coordination

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Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027

### PROFILE

- Mechanical master's student with laser cutting, CNC exposure, 3D printing workflow and CAD-to-part evidence.
- Interested in manufacturing, production support and site/project engineering roles.
- Project screenshots and source links: <https://zxie588-alt.github.io>

### EDUCATION

#### University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods. Relevant coursework: Manufacturing Processes, CNC Machining, Design for Manufacture, Manufacturing and Industrial Systems

#### Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

### PRACTICAL AND TEAM EXPERIENCE

- Manufacturing workshop training: prepared AutoCAD/DXF laser-cut profiles and compared drawings with cut parts.
- Engineering practice: interpreted force-displacement tests, controlled a robotic arm and led a small automotive-structure walkthrough.
- Technical communication: former student news lead with photography and project-documentation experience.

### PROJECT WORK

#### Manufacturing Workshop Training & DXF Preparation

*Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow*

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

#### Engineering Practice and Automotive Structure Training

*Mechanics testing, robotic arm control, automotive structure walkthrough*

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Led a small automotive-structure walkthrough covering seat/interior systems and vehicle context.

#### Ruggedized Outdoor IoT Sensor Module Design Package

*SolidWorks, enclosure design, DFM notes, Keil C, validation workflow*

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Screened corner-load, strap-lug and screw-boss regions with an FEA-style check.
- Built Keil C firmware structure with sensor, battery and state-machine modules.

#### Small Residential MVHR / ERV Revit MEP Coordination

*Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes*

- Built a NZ residential MVHR/ERV package with Revit MEP, CAD evidence and review notes.
- Modelled MVHR unit, ducts, terminals and hangers in SolidWorks; exported STEP, STL, DXF and PNG.
- Created a Revit 2026 model with plan, 3D view, airflow schedule, sheets and simple clash notes.
- Checked 60 L/s supply, 45 L/s extract and 85 L/s boost against duct velocity and pressure allowances.
- Added M-102 pressure review, BIM issue log and Mitsubishi Lossnay LGH-35RVX3-E selection note.
- Linked the design to NZBC G4/H1, Healthy Homes context and a commissioning test plan.

#### Team Systems Engineering Feasibility Study

*Stakeholders, requirements, ConOps, trade studies, risk, verification*

- Led systems-engineering structure for a 5-person technical feasibility study.
- Built stakeholder map, requirements hierarchy, ConOps, trade criteria and risk register.
- Used verification gates to keep orbit, power, thermal, communications and cost assumptions aligned.

### TECHNICAL SKILLS

- Proficient in: drawing-to-part thinking, AutoCAD/DXF laser-cut preparation, SolidWorks assemblies, STEP/STL export and component documentation
- Familiar with: CNC machining workflow, workholding, cutting fluid, chip formation, 3D printing workflow and basic workshop safety awareness
- Project support: technician/workshop communication, on-site task support, risk tracking, requirements and verification workflow